Peer Review Report

Paper Title: THE DETECT LAW: Zero-Shot Detection of Machine-Generated Text with an Integer-Exact, Heterogeneity-Aware Sequential Test
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Peer Review by Folarera Kassim (FolatheDuckofDuckingburg)

Review Metric	Evaluation
Recommendation	**Accept**
Novelty & Originality	High
Technical Rigor	High
Empirical Validation	Strong (RAID Benchmark)
Reproducibility	Outstanding (Branchless Integer C Code & ...

1. Summary of the Work

The paper introduces THE DETECT LAW, a zero-shot detector for machine-generated text that combines three per-token signals (surprise, contrast, and repetition) extracted from two small open language models (Llama-3.2–1B and its instruction-tuned twin) with an exact 32–bit integer sequential decision rule.

Key methodological contributions include:

- **Three Complementary Arms:** Measuring token predictability (surprise), chat-model preference (contrast), and repetition penalty usage (repetition).
- **A Heterogeneity-Aware Null Model:** Incorporating a per-token bias parameter to prevent distinctive human writing styles (such as poetry) from experiencing inflated false-positive rates as text length grows.
- **Sequential Decision Rule:** Utilizing an end-of-text Gaussian test alongside anytime-valid stopping bounds derived from Ville's inequality and Bennett-type bounds, bounded by a predefined 1% false-positive budget.
- **Integer-Exact Implementation:** Implementing the decision logic in roughly 325 lines of branchless 32–bit integer C code that exhibits zero divergence from a real-valued oracle.

Evaluated on the official RAID benchmark, THE DETECT LAW achieves a True Positive Rate (TPR) of **92.33% at a 5% False Positive Rate (FPR)** without attacks (86.54% at 1%) and **88.99% with attacks**, outperforming

previous zero-shot detectors such as *Binoculars* and *DivEye* while operating efficiently on a single NVIDIA A30 GPU.

2. Strengths

- **State-of-the-Art Zero-Shot Performance:** On the RAID benchmark, the detector surpasses existing zero-shot methods across all eight domains, bridging much of the historical gap between zero-shot approaches and heavily trained classifiers.
- **Novel Repetition Signal:** Traditional likelihood-based zero-shot detectors struggle with text generated using repetition penalties. The introduction of the repetition arm effectively addresses this blind spot, raising detection rates for penalty-decoded text significantly (e.g., jumping from 52.46% to 98.64%).
- **Robust Statistical Foundation:** The inclusion of a heterogeneity-aware null model solves a critical failure mode where long-form or stylized human texts (like poetry) accumulate false positives under homogeneous assumptions.
- **Computational Efficiency & Engineering Rigor:** Using small 1.24B-parameter models combined with an integer-exact C implementation ensures high throughput (scoring the entire RAID test set in 5.6 hours on an A30 GPU) and eliminates floating-point rounding discrepancies.

3. Weaknesses and Areas for Improvement

- **Vulnerability to Advanced Text Transformations:** As documented by the author, paraphrase attacks (76.57%) and synonym substitution (80.83%) remain challenging. The fixed decision rule frequently leaves paraphrased machine text labeled as "undecided". Further discussion or mitigation strategies for semantic-preserving adversarial paraphrasing would strengthen the paper.
- **Dependence on Calibration Data:** The detector's constants are calibrated exclusively on human texts drawn from the RAID training split. While disjoint from the test set, these texts share the same underlying distribution. Evaluating transferability to out-of-domain corpora, other languages, or contemporary conversational models would provide deeper insight into its generalizability.
- **Demographic Bias Evaluation:** The author explicitly notes that text written by non-native English speakers was not evaluated. Given documented biases in language model detectors against non-native writers, addressing demographic robustness or acknowledging this as an open limitation is essential for ethical deployment.

4. Minor Comments & Technical Notes

- **Score Scaling Clarification:** Appendix A provides a helpful explanation regarding the transformation of raw integer scores into $\sqrt{\text{score}/256}$ to align with RAID's threshold search. This practical implementation detail is valuable for reproducibility.

- **Reproducibility:** The provision of exact SHA-256 hashes for the C rule and submission files, along with open-source code repositories, sets a high standard for artifact availability.

5. Final Recommendation

Accept. THE DETECT LAW presents a technically rigorous, highly efficient, and empirically strong advancement in zero-shot machine-generated text detection. The paper successfully combines statistical theory (martingales, Ville's inequality, heterogeneity-aware nulls) with practical systems engineering (branchless integer C code). Addressing the limitations regarding adversarial robustness and cross-domain calibration in future work will further broaden its impact.